INCH-POUND

MIL-M-5631C
15 January 1991
SUPERSEDING
MIL-M-5631B(ASG)
15 June 1967

MILITARY SPECIFICATION
MOUTON (FOR FLIGHT CLOTHING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mouton, a simulated fur produced from sheared sheepskin of aeronautical-clothing grade.

1.2 Classification. This shall be furnished in the following four types of pile heights and class of wool grades, as specified (see 6.2):

Type I	+ 1/2 inch
Type II	+ 5/8 inch
Type III	+ 3/4 inch
Type IV	+ 1 inch
Class 1	+ 58's grade of wool
Class 2	+ 64's grade of wool

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-T-60 Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Technology & Industrial Services Division, SA-ALC/TIRDM, Kelly AFB, TX 78241-5609 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8330

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NN-P-530	Plywood, Flat Pane.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
QQ-S-781	Steel, Strapping, Flat.
PPP-B-1055	Barrier Material, Water Proofed Flexible.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements.

STANDARDS

FEDERAL

*	FED-STD-191	Textile Test Methods.
*	FED-STD-311	Leather Methods of Sampling and Testing.

* (Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3992-85	Fineness of Wool Top or Mohair Top and Assignment of Grade.
ASTM D 3951	Standard Practice for Commercial Packaging.

(Application for copies of the above publication should be addressed to the American Society for Testing and Materiels, 1916 Race Street, Philadelphia, PA 19103).

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. The skins used for this materiel shall be sheepskin packer's takeoff, or equivalent quality. The quality or grade of the wool shall be not lower than that specified by the procuring activity, and shall be in accordance with the official standards for grades of wool tops as issued by the U. S. Department of Agriculture.

3.2 Shape and trim. The desired shape and allowable trim of the finished skin shall be in accordance with figure 1. Necks, flanks, and butts shall be trimmed when the finished fiber does not extend to the edge, or when wool density or skin defects affect the quality of the finished skin. The skins shall be so trimmed as to avoid unreasonable cutting losses. The footage allowance shall be in accordance with standard commercial practice for this class of goods.

3.3 Cuttable area. The cuttable area of each skin shall be not less than 5 square feet.

3.4 Defects.

3.4.1 Defects acceptable without repair 1/. Minor imperfections in the leather or the wool, which do not constitute either a hole or an obvious bare spot, are acceptable. Fifty percent of the skins on any order may contain up to three bare spots not to exceed 1-1/2 inches in greatest dimension, provided that the spots occur within 8 inches of the edge of the skin. Twenty percent of the above 50 percent may contain not more than two holes in each skin, not greater than 1-1/2 inches in greatest dimension, provided that the holes occur within 8 inches of the edge of the skin; or each skin may contain a single brand mark, hard spot, hole, or bald spot not more than 4 inches in greatest dimension, so located that it will not interfere with patterning, and which constitutes the only defect in the skin. An allowance shall be made by the manufacturer for the loss in cuttable area owing to the presence of bare spots and holes.

1/ This requirement is applicable only to skins supplied for qualification tests of a manufacturer's product and to skins purchased or furnished to the Government.

3.4.2 Defects repairable. Grain cracks and defects which are not acceptable shall be repaired by the process known as serging, except that there shall be not more than three seams in a skin, not more than one of which shall extend from edge to edge, and not more than one-third of the skins presented for acceptance shall be repaired skins.

3.4.2.1 Serging. Serging shall be accomplished by fur sewing on an overstitch machine, and the finished seam shall be flat and shall not interfere with pattern cutting.

3.5 Tannage. The skins, with wool attached, shall be properly and fully tanned by chrome tannage method.

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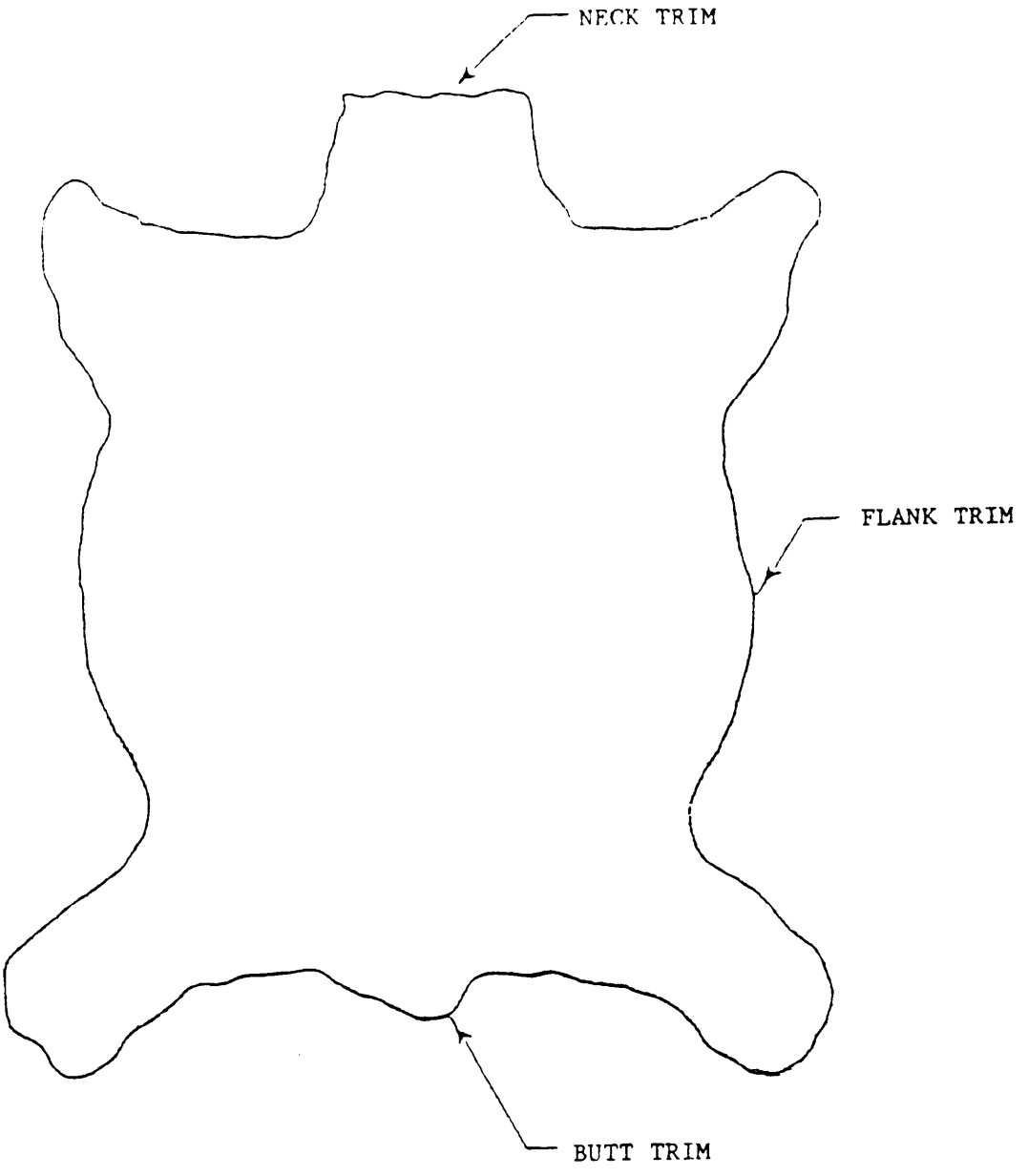


FIGURE 1. Desired shape of finished skin.

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3.6 Leather finish. The flesh side of the skin shall be uniformly buffed to provide an even suede surface. The leather, before proofing, shall be uncolored, except for the color which may be absorbed in tanning the skin and in dyeing the wool.

3.7 Wool finish. The wool fiber shall be chemically treated and processed to produce a weatherproof finish. The wool fibers shall show no shortening, dulling, kinking, or change in surface appearance when subjected to the weatherproofness test specified in 4.5.7. The wool fibers shall be processed in such manner that there shall be no separation of the fibers as to cause a continuous line of open fibers when the skin is folded 180 degrees upon itself, with the wool fibers on the outside.

3.8 Wool fiber. The fiber length of the wool shall be 1/2 inch, 5/8 inch, 3/4 inch, or 1 inch, all ±1/16 inch, as specified by the procuring activity, and the pile shall be uniform throughout the skin.

3.9 Color. Unless otherwise specified, the wool fiber shall be uniformly and evenly dyed with a fast-color dye to match either brown, munsell color notation, hue 10R to 5YR, value 2 to 3, chroma 0.2 to 2, or USAF Sage Green 523.

3.10 Chemical requirements. The sheared leather shall conform to the chemical requirements specified in Table I.

TABLE I. Chemical requirements.

Tests	Minimum	Maximum
Organic solvent extract (grease), moisture-free basis (percent)	10	22
Ash, total (percent) 1/	--	8.0
Acidity (ph) 2/	4.0	6.0

1/ On a grease-free, moisture-free basis.
2/ On a grease-free basis.

3.11 Physical requirements.

3.11.1 Colorfastness. The color of the dyed wool shall show no signs of crocking or color transfer when rubbed with a dry or damp cloth.

3.11.2 Breaking strength. The average breaking strength of the sheared leather shall be 30 pounds with no individual test showing less than 25 pounds.

3.11.3 Seam strength. The breaking strength of seams shall be not less than 50 percent of the breaking strength of the leather adjacent to the seamed area.

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3.11.4 Curl. The finished skins shall not curl or be appreciably affected when subjected to the curl test specified in 4.5.6.

3.11.5 Shrinkage temperature. The leather shall not shrink when exposed to a temperature of 85° C.

3.12 Identification of product.

3.12.1 Tagging. Each skin shall be tagged for identification. The identification data shall be in accordance with MIL-STD-130.

3.12.2 Marking on skin. In addition, each skin shall be marked or stamped near one edge of the flesh side with the specification number, the applicable type, class, and color of mouton.

3.12.3 Mothproofing. The mouton shall be treated with a mothproofing compound so that the wool fiber shall contain a minimum of 0.5 percent of flourine determined on a moisture-and-fat-free basis. Mothproofing compounds shall contain no arsenic or other chemicals that will promote skin irritation or dermatitits.

3.13 Workmanship. All details of workmanship, including selection of skins, tanning, dyeing, and finishing, shall be in accordance with high-grade commercial practice. The finished skin shall be free from grain cracks, loose open wool, greasy leather, open ribs, and other defects which may affect the appearance or serviceability, or substantially increase the cutting losses.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. All inspections specified herein are classified as quality conformance inspections for which necessary sampling techniques and methods of testing are specified herein.

4.3 Test conditions. All materials and samples shall be conditioned at standard controlled temperature and humidity until equilibrium has been attained, prior to testing. Temperatures shall be maintained at 70° ±5° F (21° ±3° C) and 65 percent ±2 percent relative humidity.

4.4 Sampling tests.

4.4.1 Sampling. Samples shall be selected in accordance with FED-STD-311 from each delivery of skins of one type and class submitted for acceptance at one time (see 4.4.1.1), and subjected to all the inspections of this specification. A unit of delivery shall be one skin.

4.4.1.1 Selection of specimens. Specimens for physical requirements (see 3.11) and chemical tests specified in 4.5.8 shall be selected from the areas of the hide as indicated in FED-STD-311.

4.4.1.2 Rejection and retest. Failure of any mouton skin to conform to any one of the requirements of this specification shall be cause for rejection of the lot represented. Skins which have been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Skins rejected after retest shall not be resubmitted without specific approval of the procuring activity.

4.5 Inspection methods.

4.5.1 Examination of product. Each skin shall be examined for conformance with type, shape, finish, color, defect, marking, and workmanship requirements.

4.5.2 Grade of wool. The fineness of the wool fiber shall be determined by the width method and the use of a microscope in accordance with ASTM D3992-85, except that a minimum of 300 fibers shall be required per test. The sample of wool fiber used for wool grading shall be taken from the area located at the center of the back of the skin midway between the neck and tail.

4.5.3 Colorfastness. Resistance to crocking (colorfastness) shall be accomplished in accordance with method 3011 as specified in FED-STD-311.

4.5.4 Breaking strength. The breaking strength test shall be in accordance with method 2011 specified in FED-STD-311, except that the specimen shall be cut with long dimension in the backbone-to-belly direction. The tensile strength shall be reported as the average of the five test specimens.

4.5.5 Seam strength. Five 6- by 6-inch specimens shall be taken across a seam so that the seam is approximately centered on the specimens, and five specimens adjacent to the seam and in the same direction as the seam-test specimens shall likewise by taken. All 10 specimens shall be tested in accord-

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ance with method 5100 specified in FED-STD-191 and the strength of the seamed specimens computed as a percentage of the strength of the unseamed specimens.

4.5.6 Curl. A sample of leather 2 inches square shall be cut and the wool sheared off in any suitable manner that avoids damaging the skin, after which the specimens shall be placed in a beaker containing 400 milliliters (ml.) of distilled water. The water shall be heated to a temperature of 149° ±2° F (65° ±1° C) and the prepared specimen immersed for 3 minutes. The sample shall then be removed, air dried at room temperature, and examined for flatness.

4.5.7 Weatherproofness. Water at room temperature shall be sprinkled on the wool side of the mouton under test. The drops shall be allowed to remain for 15 minutes, then the samples shall be shaken to remove the excess water and allowed to dry thoroughly until there is no feeling of dampness apparent to the touch. The drying time shall be not less than 5 hours in still air, nor less than 1 hour in circulating air under normal atmospheric conditions.

4.5.8 Chemical tests. Unless otherwise specified herein, test conditions and methods for chemical tests shall be in accordance with the following methods specified in FED-STD-311.

Tests	Method No.
Organic Solvent extract (grease)	1/ 6321
Ash, total	6421
Acidity (pH)	6621

1/ Chloroform shall be used as the solvent instead of the petroleum ether, in grease content determination.

4.5.9 Mothproof (fluorine test).

4.5.9.1 The chemicals, reagents, and indicator used shall be as follows:

Chemicals and reagents:
50 percent H_2SO_4 in distilled water (by volume)
N/100 Th $(NO_3)_4$ solution
N/10 HCl solution
10 percent NaOH solution
Indicator: 0.05 percent aqueous solution alizarin red.

4.5.9.2 Procedure. A sample of the wool yarn weighing approximately 5 grams shall be extracted in a soxholet extractor with either carbon tetrachloride, Stoddard solvent, or tetrachloroethylene for a period of 4 hours. After extracting, the solvent should be evaporated from the specimen and then dried to constant weight in an oven at a temperature of 221° ±3° F (105° ±1° C). The dried sample shall be placed in a three necked distilling flask with 50 ml. of 1 to 1 sulfuric acid. A magnetic stirring bar shall be added and the adapter shall be immersed in 20 ml. of distilled water in receiving beaker to insure complete solution of gases. An infrared lamp shall be so mounted

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that its rays are focused on the surface of the liquid in the flask. Heat shall be applied gradually with the variable transformer and the stirring bar set in motion. After the materiel is completely dissolved, the heat shall be increased until distillation begins. Liquid temperature shall be maintained between 275° and 284° F (135° to 140°C) begins. Liquid temperature shall be maintained between 275° and 284° F (135° to 140°C) by dropwise addition of distilled water from the separatory funnel. Heat and water feed shall be adjusted to produce a rate of distillation of approximately 1 drop per second. The solution shall be distilled until approximately 300 ml. of distillate has been collected. A volume of exactly 500 ml. shall be made up in a volumetric flask by the addition of distilled water. A 100-ml. aliquot shall be transferred from the volumetric flask to a titration flask and 0.5 ml. of alizarin monosulphonate indicator added. It shall be neutralized to a violet color with 10 percent sodium hydroxide (see Note 1). Hydrocholoric acid (0.1N) shall be added until the pale yellow color is restored and then added in excess (see Note 2). The solution shall be titrated with 0.01N thorium nitrate from a microburst to the appearance of a faint permanent pink coloration. Percent of flourine shall be computed by means of the following formula:

$$\text{Percent flourine} = \frac{A \times .095}{W}$$

Where: A = cc of Th (NO3)4

W = weight of dried sample

Note 1: Twenty-ml. aliquot of distillate should never require more than 1 drop 10 percent NaOH solution of neutralization. When more is required, contamination of the distillate by decomposition products (formed by localized overheating during distillation) is indicated. These products interfere with the titration.

Note 2: The pH of the solution taken for titration is important. The pH provided by the addition of 5 drops excess N/10 HCl is ideal (i.e., pH 3.2). The use of buffers should be avoided as they will obscure the end point even further.

Note 3: Sodium ion concentration in the distillate should be kept to a minimum since these ions cause erroneous (high) results.

4.5.10 Color matching. Dyed mouton shall be matched to the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7,500° K and shall be a good approximation to the standard shade under incandescent lamplight at 2,800°K.

4.6 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance with section 5.

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5. PACKAGING

5.1 Preservation and packaging. Packaging shall be Level A, C or Commercial as specified (see 6.2).

5.1.1 Level A. Mouton skins of one type and class shall be laid flat, faced wool to wool, in accordance with the schedule described in 5.1.1.1, in full telescopic design boxes, style FTC, conforming to PPP-B-636 and having a minimum Mullen test of 350 pounds. The design of the telescopic box shall be securely fastened with a minimum of three flat steel straps girthwise, and a minimum of two flat steel straps lengthwise of the box. Strapping shall conform to QQ-S-781 and shall be of a minimum size of 1/2 inch by 0.015 inch.

5.1.1.1 Packaging schedule (level A).

Type of skin	Number of skins per box
I	72
II	60
III	60
IV	48

5.1.2 Level C. Mouton skins of one type and class shall be separately packaged in accordance with the applicable requirements of MIL-STD-2073-1.

5.1.3 Commercial. Mouton skins of one type and class shall be separately packaged in accordance with the applicable requirements of ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or Commercial as specified (see 6.2).

5.2.1 Level A. Mouton skins, packaged as described in 5.1, shall be packed in cleated-plywood or nailed wood boxes conforming to PPP-B-601 (overseas type) and PPP-B-621, respectively. Plywood, if used, shall be of type I or II, class 2 of NN-P-530. Unless otherwise specified, boxes shall be furnished with a sealed case liner fabricated from barrier material conforming to PPP-B-1055. All seams and closures shall be completely sealed with adhesive conforming to class 1 of PPP-T-60. The width of the mating surfaces of all seams shall be not less than 3/4 inch. Boxes shall be closed and strapped in conformance with the appendix of the applicable specification. The gross weight of shipping containers shall not exceed approximately 200 pounds.

5.2.2 Level B. Mouton skins, packaged as specified in 5.1, shall be overpacked in wood-cleated plywood boxes, nailed wood boxes, or corrugated fiber boxes conforming to PPP-B-601, PPP-B-621 and PPP-B-636, weather resistant. As far as practical, exterior shipping containers shall be of uniform shape and size and contain identical quantities. Fiberboard boxes shall conform to the special requirements of the applicable specification. The gross weight of wood and wood-cleated boxes shall not exceed 500 pounds. Boxes shall be closed in accordance with the appendix of the applicable specification.

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f dimensional characteristics. 4.2.2.2 Examination to

5.2.3 Level C. Mouton skins, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-STD-2073-1.

5.2.4 Commercial. Mouton skins, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of ASTM D 3951.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified in the purchase document 1/
MOUTON (FOR FLIGHT CLOTHING)
Type (I, II, III, or IV, as applicable)
Class (1 or 2, as applicable)
Total square feet (minus allowances)
Specification MIL-M-5631
Contract or Order No.
Date of Manufacture
Quantity of skins
Name of manufacturer
Name of contractor (if different from manufacturer)

1/ The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No. or other identification as provided by the procuring activity.

5.3.1 Commercial. Mouton skins shall be marked in accordance with the applicable requirements of ASTM D 3951.

5.4 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order shall be palletized in accordance with MIL-STD-147, the pallet pattern shall first be approved by the contracting officer.

6. NOTES

6.1 Intended use. The mouton covered by this specification is intended for use in the manufacture of flight clothing.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and color of mouton required (see 1.2.).

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c. Both the level of packaging and the level of packing required (see 5.1 and 5.2).

d. When palletization is required (see 5.4).

6.3 Standard sample (applicable to sage green, no. 523). For access to standard samples, address the procuring activity issuing the invitation for bids (see 3.9).

6.4 Subject term key word listing.

Wool
Mohair
Artificial wool
Artificial mohair
Artificial fur

6.5 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - AS
Air Force - 99

Reviewer activities:
DLA - CT

User activity:
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Preparing activity:
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